NIRANJAN SHRESTHA

Power Electronics Design Engineer — Power Conversion, Digital Control & Hardware Validation

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PROFESSIONAL SUMMARY

Power electronics design engineer with 6+ years of industry and industry-partnered R&D experience across AC/DC, DC/AC and DC/DC conversion: boost PFC rectifiers, SiC MOSFET full-bridge inverters, active/passive rectifiers, and resonant DC–DC converters up to 7.7 kW. Owns the full design cycle - simulation and modeling (MATLAB/Simulink, Simscape, PLECS, LTspice), magnetics and compensation design, digital control firmware on TI C2000 and STM32, PCB design, and hardware validation test planning against SAE, CSA, UL and IEEE 519 requirements.

Delivered **95% DC–DC efficiency** on 3.7 kW platforms and **93% end-to-end efficiency at 3.6% source-current THD**. Currently conducting applied research and development on novel system architecture, advanced converter topologies, and control methods for EV charging up to 7.7 kW at Wireless PnC Inc., with responsibility for identifying research directions, prototyping, hardware testing/validation mentoring junior engineers, and translating findings into manufacturable designs compliant with SAE, CSA, and UL automotive standards. Seeking to apply converter design and control research to the next generation of charging infrastructure.

TECHNICAL COMPETENCIES

- **Converter topologies:** AC/DC boost PFC rectifiers · DC/AC two-level full-bridge and half-bridge inverters (SiC MOSFET) · DC/DC resonant converters (LLC, series–series, LCC–LCC, hybrid compensation) · synchronous and diode rectification · soft switching (ZVS/ZCS)
- **Modeling & simulation:** MATLAB/Simulink · Simscape · PLECS · LTspice · M-language · circuit- and system-level co-simulation · model-to-hardware correlation within 5%
- **Control & modulation:** Phase-shift and frequency control · Duty cycle control · CC/CV charging control · closed-loop design · small-signal modeling and stability analysis · load-transient and misalignment-tolerant control
- **Magnetics:** Resonant coil design and characterization (circular, rectangular, hexagonal) · compensation network design · coupling and misalignment analysis
- **Embedded systems & firmware:** TI C2000 · STM32 · C/C++ · Python · digital control loop implementation · SPI and UART communication
- **Hardware design & test:** KiCad schematic design and PCB layout · board fabrication and bring-up · thermal and efficiency characterization · power-quality (THD) measurement · fault injection and root-cause analysis · bench and vehicle-level testing · AutoCAD design for building electrification
- **Real-time validation:** Simulink-to-embedded deployment via automated code generation · introductory OPAL-RT RT-LAB exposure
- **Standards & compliance:** Validation test plan authoring · SAE, CSA and UL wireless charging standards · IEEE 519 harmonic compliance · BS 7671 test and inspection · vendor test-report audits against standards · technical specifications and design documentation
- **Energy storage & renewables:** Battery state-of-health estimation for BMS · V2G-capable charging architectures · solar PV and storage sizing · distributed-generation interconnection applications

PROFESSIONAL EXPERIENCE

Electrical Engineer — Power Conversion and Control R&D for Wireless Charging System — May 2024 – Present

Wireless PnC Inc. · Victoria, British Columbia

- Own end-to-end design, dynamic modeling, circuit- and system-level simulation, control design and hardware validation of wireless charging systems up to 7.7 kW across EV platforms, delivering 95% DC–DC efficiency.
- Design and fabricate AC/DC and DC/AC power stages, including boost PFC rectifiers, rectifiers, and full-bridge inverters, achieving 93% end-to-end efficiency and 3.6% source-current THD under non-linear loading, evaluated against IEEE 519 and automotive power-quality requirements.
- Perform KiCad schematic design, PCB layout, fabrication and hardware bring-up for 8 converter boards; author technical specifications, detailed schematics and design documentation.
- Author validation test plans for all project phases and integrate evolving SAE, CSA and UL requirements into design practice; execute bench and vehicle-level testing and report findings to engineering leadership.
- Evaluate emerging converter topologies and control schemes against product roadmaps; advanced 3 concepts to working prototypes for e-bike and e-scooter WPT systems.
- Resolve cross-domain integration issues with hardware, firmware and mechanical engineering through joint root-cause analysis.
- Mentor and supervise 4 electrical interns across simulation, hardware design and validation, establishing reusable modeling and test-procedure frameworks now applied across 7.7 kW WPT projects.

Graduate Research Assistant — Power Electronics and Control for Wireless Charging — Jan 2022 – Apr 2024

Ontario Tech University · Oshawa, Ontario

- Delivered complete R&D of a resonant wireless EV charging system achieving 92.5% peak DC–DC efficiency at 3.7 kW, funded by NSERC and MITACS with an industry partner.

- Developed a novel hybrid compensation network paired with phase-shift control for WPT system - CC/CV charging under misaligned conditions, the basis of three first-author IEEE conference papers (ITEC 2023, ITEC 2024, APEC 2025).
- Built and experimentally verified dynamic models of DC–DC converters, inverters, rectifiers and resonant converters in MATLAB/Simulink, Simscape and PLECS, correlating simulated efficiency and transient response to hardware within 5%.
- Validated closed-loop control by deploying Simulink/Simscape models to embedded targets during system integration and testing.
- Characterized 3 coil geometries and 3 compensation configurations, quantifying efficiency and power-transfer sensitivity to misalignment and air-gap variation.

Electrical Engineering Intern — Distributed Generation **Jun 2022 – Apr 2023**
Oshawa Power · Oshawa, Ontario

- Audited vendor equipment test reports against applicable industry standards, identifying non-conformances before approval.
- Prepared micro-embedded generation connection applications for net-metering distributed energy resource interconnections.
- Developed regression-based forecasting models for energy production and revenue across 5 solar PV and wind assets, used in utility planning.

Project Engineer — Solar PV System Design **Nov 2019 – Dec 2021**
Sunshine Energy Pvt. Ltd. · Kathmandu, Nepal

- Site survey along with overall system design, cost estimation, and tender bid document preparation of off-grid and on-grid solar PV system, solar water pumping/irrigation system, solar home system, and solar street lighting system projects, totalling nearly 60 kW.
- Translated site-survey constraints into generation capacity, battery storage sizing and runtime design decisions.

Electronics Technician — Electrical Test & Maintenance **Feb 2018 – Aug 2019**
Defence Infrastructure Organisation – British Gurkhas Nepal · Lalitpur, Nepal

- Diagnosed and maintained electrical equipment like generators, motors, boilers and solar thermal systems; executed electrical installation testing and inspection to BS 7671:2008 with compliant certification documentation.

EDUCATION

Master of Applied Science (MASc), Electrical & Computer Engineering **Jan 2022 – Apr 2024**
Ontario Tech University · Oshawa, Ontario

Thesis: Hybrid Compensation-based Misalignment-Tolerant Wireless Power Transfer For E-Mobility (3.7 kW, CC/CV phase-shift control). Supervisor: Dr. Sheldon Williamson. Funding: NSERC, MITACS Accelerate.

Bachelor of Engineering, Electrical Engineering **Nov 2013 – Sep 2017**
Tribhuvan University, Pulchowk Campus · Lalitpur, Nepal

PUBLICATIONS

9 peer-reviewed publications (1 journal article, 1 book chapter, 7 IEEE conference papers) · 55 citations · h-index 5

- Y. Shanmugam, R. Narayanamoorthi, V. K. Ramachandaramurthy, P. Bernat, et al., "Machine learning based optimal design of on-road charging lane for smart cities applications," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 4, 2024.
- N. Shrestha, J. Son, and S. Williamson, "Smart charging of electric vehicles," in *Battery Technologies for E-Mobility*, 2025, pp. 289–337.
- N. Shrestha, V. S. R. V. Oruganti, and S. Williamson, "Digitally controlled misalignment-tolerant inductive power transfer system with adaptive hybrid compensation for CC/CV charging of e-scooter," in *Proc. IEEE APEC*, 2025, pp. 801–808.
- N. Shrestha and S. Williamson, "Hybrid compensation network for misalignment-tolerant constant current/constant voltage charging for wireless power transfer system," in *Proc. IEEE ITEC*, 2024, pp. 1–6.
- J. Son, N. Shrestha, and S. Williamson, "Comprehensive comparative analysis of circular, rectangular, and hexagonal coils for wireless charging of e-mobility," in *Proc. IEEE APEC*, 2024, pp. 2915–2921.
- N. Shrestha, J. Son, and S. Williamson, "Constant current/constant voltage charging via series-series compensated resonant inductive wireless charging for electric vehicle," in *Proc. IEEE ITEC*, 2023, pp. 1–6.
- N. Shrestha, A. Samanta, F. C. Fietosa, and S. Williamson, "State-of-the-art wireless charging systems for e-bikes: Technologies and applications," in *Proc. IEEE PEDS*, 2023.
- A. Samanta, A. Huynh, N. Shrestha, and S. Williamson, "Combined data driven and online impedance measurement-based lithium-ion battery state of health estimation for electric vehicle battery management systems," in *Proc. IEEE APEC*, 2023, pp. 862–866.
- K. T. Lulbadda, R. De Seram, N. Shrestha, T. Sidhu, and S. S. Williamson, "An analysis of advanced soft-switching techniques for dc–ac power converters based on auxiliary circuits," in *Proc. IEEE ICIT*, 2023, pp. 1–6.

AWARDS & PROFESSIONAL ACTIVITY

Invited tutorial co-presenter, IEEE WPTCE 2026, Halifax (with S. Williamson and V. Oruganti): SiC converter-based EV charging architectures with V2G capability

IEEE ITEC Student Travel Award (2023) · IEEE ITEC Coil Winding Competition, 1st Runner-Up (2023) · IEEE SPEC Scholarship Award (2022) · Active Volunteer, IEEE Toronto Durham Region Sub-section (May 2022 – Apr 2024)

REFERENCE

Available upon request